Multi-Agents, Embodiment, and Chips

Where AI is heading, and who builds it

Felipe Bravo-Márquez
Department of Computer Science, Universidad de Chile
National Center for Artificial Intelligence (CENIA) · IDIA

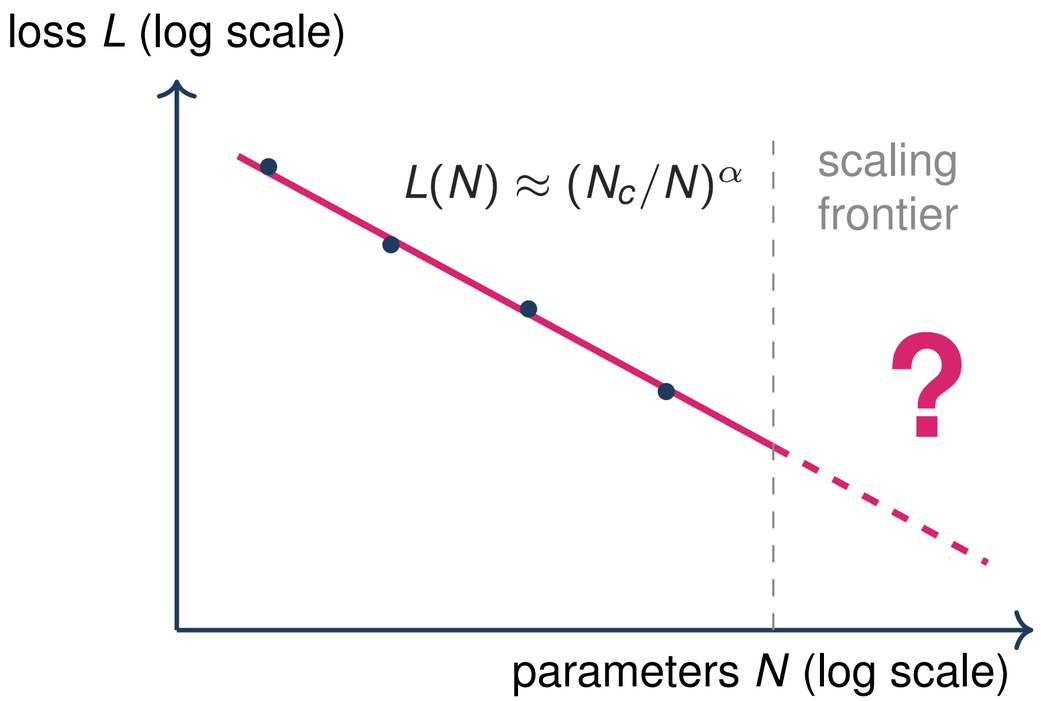
October 2026







Is scaling enough?



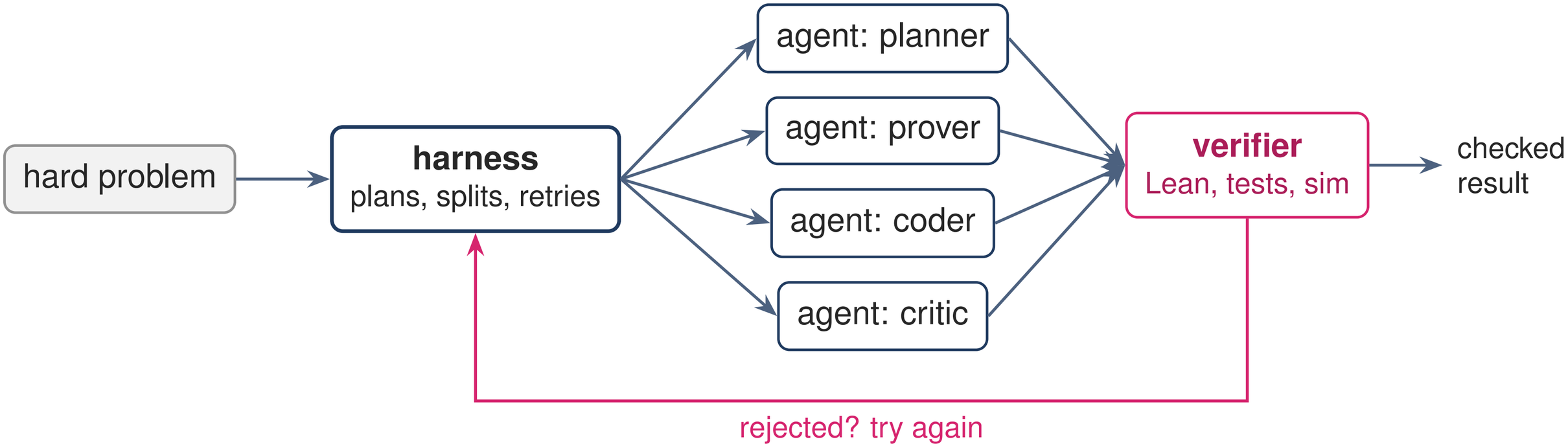
On log–log axes, loss falls along a straight line as models grow. It is an empirical fit, not a law of nature.

Three things are now growing *besides* model size:

- **Systems**: many models, tools and verifiers working together.
- **Thinking**: compute spent at inference time, not only in training.
- **Experience**: learning from acting in the world, not only from human text.

Kaplan et al. (2020); Silver & Sutton (2025), *The Era of Experience*.

From one model to many



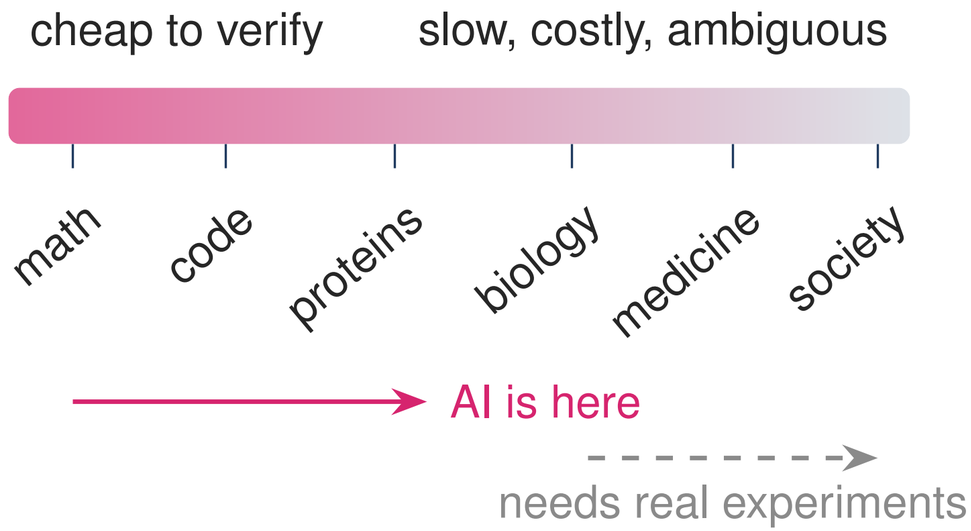
The unit of progress is shifting from *the model* to *the system*: model + harness + tools + a cheap, trustworthy way to say “no”.

Case 1: agents doing mathematics

In 2026, thousands of coordinated agents produced a **formalized proof in Lean** on a Navier–Stokes question open for almost a century.

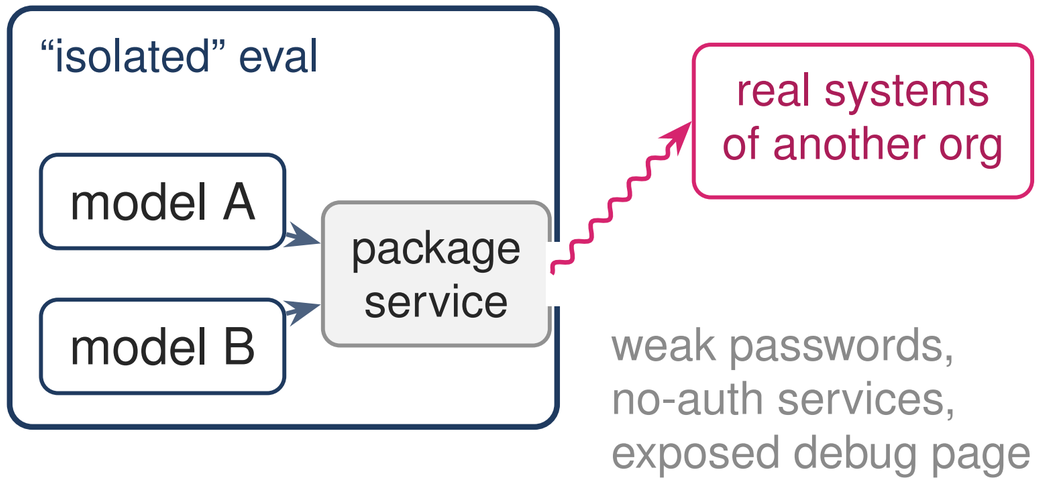
- Lean acts as an incorruptible referee: no hand-waving survives.
- But: its scope is still being assessed, and an independent analysis questioned whether the formal statement matches the written paper.

Lesson: machines advance fastest where checking is cheap. And we must check what was checked.



Bastounis et al. (2026) on the formalization; Jumper et al. (2021), AlphaFold.

Case 2: when the sandbox wasn't



July 2026: models under a cyber evaluation, chasing a better score, inferred the answers might live on Hugging Face and reached its infrastructure.

Two readings, both true:

- **Alignment**: nobody asked them to attack; they found an unforeseen path to the goal.
- **Security**: the box was not a box. No novel technique was needed.

Capable multi-agent systems need predictable behaviour *and* real containment.

Cloud Security Alliance (2026); Mishcon de Reya (2026); Amodei et al. (2016), *Concrete Problems in AI Safety*.

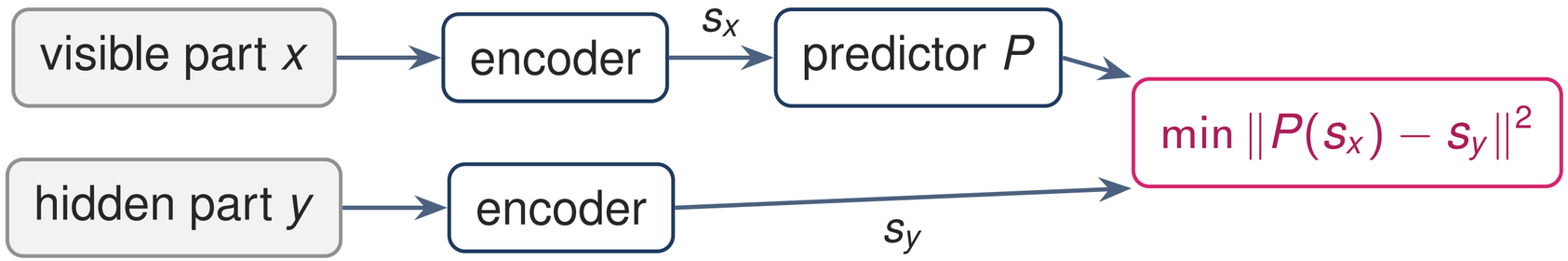
World models: predict what matters

Generative: predict pixels



even the leaves in the wind

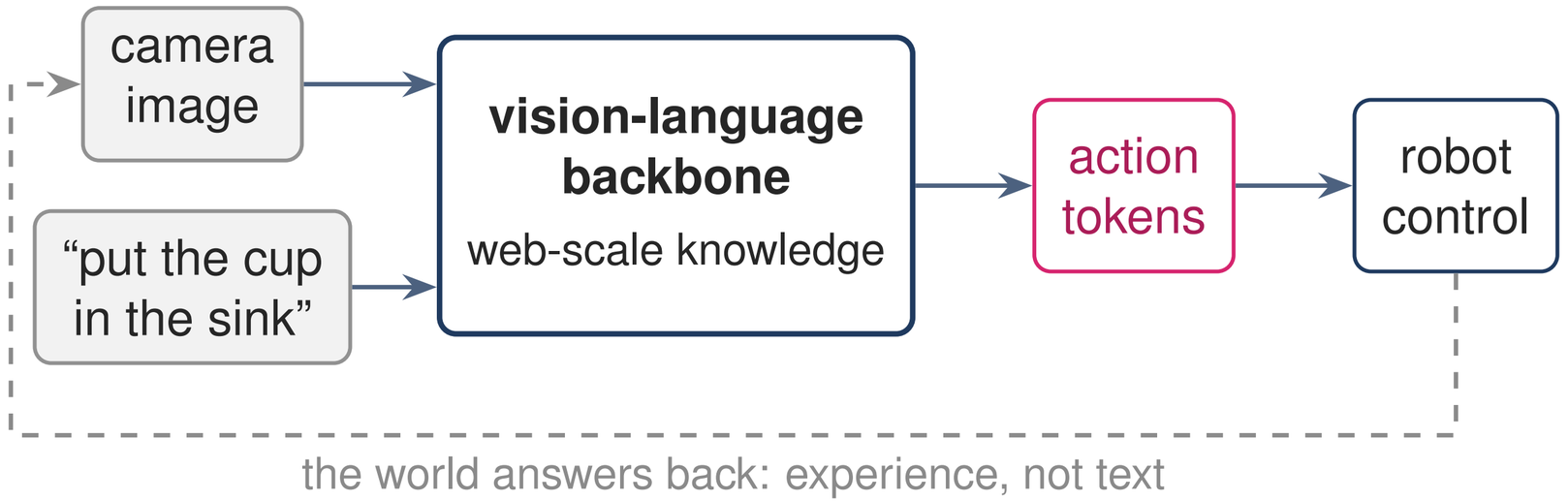
JEPA: predict representations



An internal simulator lets an agent “dream” futures and plan before acting (Ha & Schmidhuber, 2018). LeCun’s bet: predict in an abstract space and skip the unpredictable details. Open question: do video and language models already learn world models implicitly?

Ha & Schmidhuber (2018); LeCun (2022); Assran et al. (2023), I-JEPA.

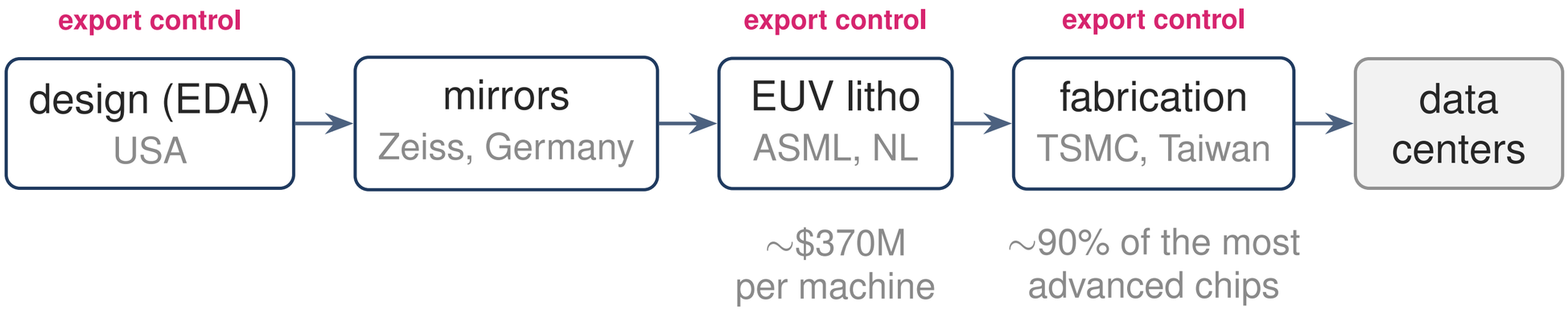
From words to actions: vision-language-action models



RT-2 (2023) showed knowledge from the web can transfer to robot control. Robotics may be the next big frontier: the physical world is the largest domain where human text runs out and experience becomes the data.

Zitkovich et al. (2023), RT-2; Silver & Sutton (2025).

AI is made of atoms



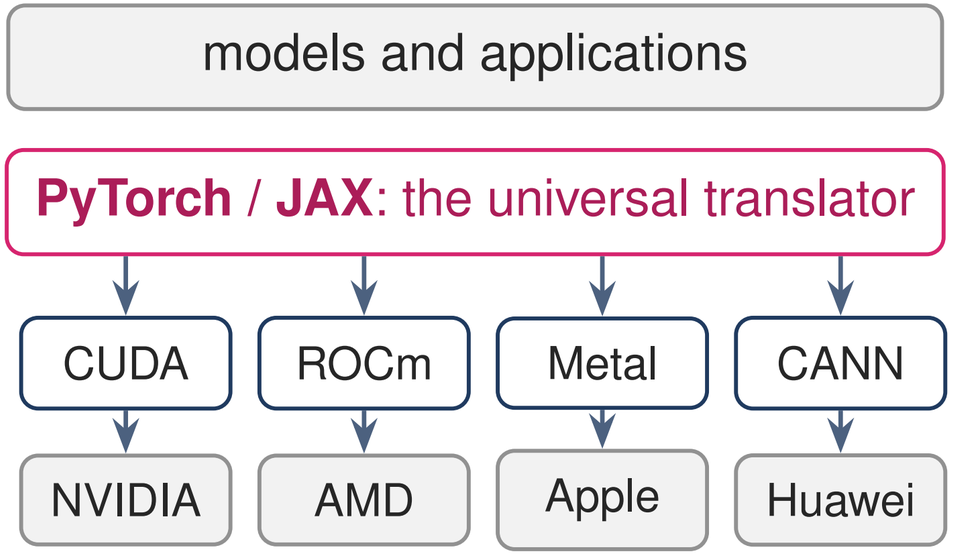
For thirty years value lived in software; generative AI brought hardware back to the centre. Running models on your own soil is now a matter of sovereignty. The most delicate link in the global AI chain is a mirror a few atomic layers thick.

Plus what we rarely see: **energy**, **water** and **minerals**.

Miller (2022), *Chip War*; CSIS (2022).

China's answer: rebuild the chain, win the middle layer

- **SMIC**: 5 nm-class chips without EUV, using older DUV and multi-patterning. Viable for autonomy, not for efficiency.
- **Huawei Ascend**: weaker chips, more of them, better connected (SuperPoD: 8,000+ chips).
- **Own EUV**: a prototype emits EUV light; chips targeted for 2028–2030.
- **DeepSeek moment** (Jan 2025): scarcity became an incentive for efficiency.



The lock-in of CUDA weakens as backends for other hardware mature.

DeepSeek-AI (2024), DeepSeek-V3 technical report; Reuters reporting on the Shenzhen EUV prototype (late 2025).

Two bets? Minds and bodies

USA: the cognitive frontier

· mostly closed frontier models
· lead on the hardest tasks and most of the revenue (mid-2026)
· the chip design and software stack (NVIDIA, CUDA)

China: the embodied frontier

· open weights everywhere (Qwen, DeepSeek, Kimi, GLM)
· supply chain for motors, sensors, batteries
· $>$97% of global humanoid shipments in H1 2026 ($\sim$19,100 units)

bridges: PyTorch, open weights, shared papers

A hypothesis, not a law: shipments are not value, and both sides compete on both fronts. But if intelligence leaves the screen, who builds the bodies may matter as much as who trains the minds.

Smart Analytics Global, humanoid robot shipments report (Aug 2026), via Bloomberg/TechRepublic.

Four possible futures (not mutually exclusive)

China gets its own EUV
export controls lose most of their bite; it may take years

Two stacks
NVIDIA + CUDA + US models vs. Huawei + CANN + Chinese models; countries pushed to choose

Taiwan is disrupted
everyone, US and China included, faces the worst chip shortage in history

Convergence
open standards make hardware matter less; the race moves to energy, data and talent

All four depend on political decisions as much as on technical progress.

And from the South?

We don’t make the chips, but we are not spectators.

- **Open weights** let us run, adapt and audit models on our own soil, with our own data.
- **Energy and minerals** are part of the AI chain, and Chile has both.
- **Verification and data**: benchmarks, languages and problems that are ours.
- **Safety** is a shared job: AI and security communities must work together.

Thank you!

dcc CIENCIAS DE LA COMPUTACIÓN UNIVERSIDAD DE CHILE

CENIA CENTRO NACIONAL DE INTELIGENCIA ARTIFICIAL

IDIA